PROJECTION OF SOLIDS

11.2 CLASSIFICATION OF SOLIDS

Solids are usually classified as given below:

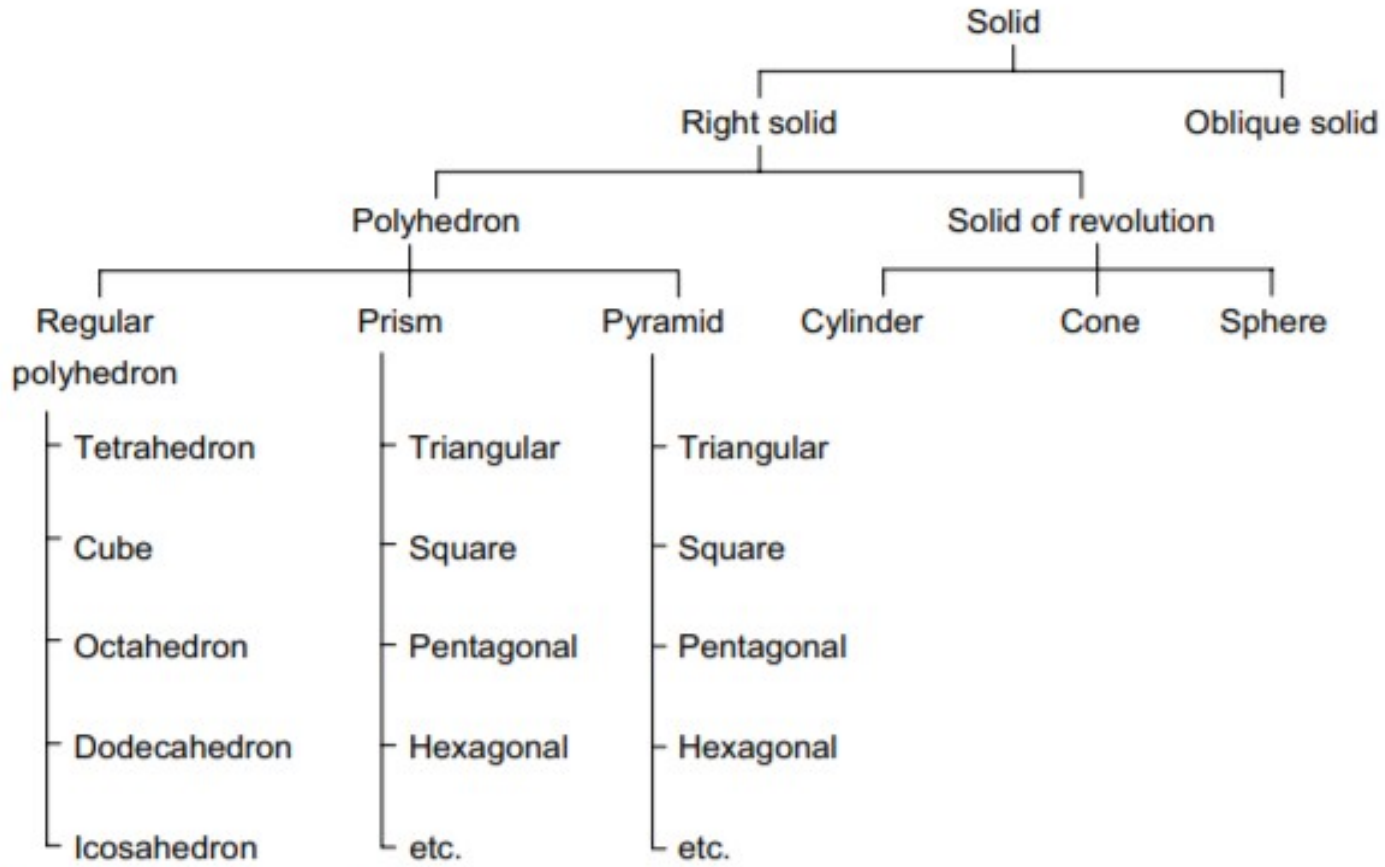


Fig. 11.1 *Classification of regular solids*

11.2.2 Prism

A prism is a polyhedron with two *n*-sided polygonal bases which are parallel and congruent, and lateral faces are rectangles. All cross-sections parallel to the bases are congruent with the bases. An imaginary line that joins the centre of the bases is called an axis. *A right and regular prism* has regular polygonal bases, axis perpendicular to the bases and all the faces are equal rectangles, as shown in Fig. 11.3. Prisms are named according to the shape of their base, so a prism with a triangular base is called a *triangular prism*; a square base is called a *square prism* and so on.

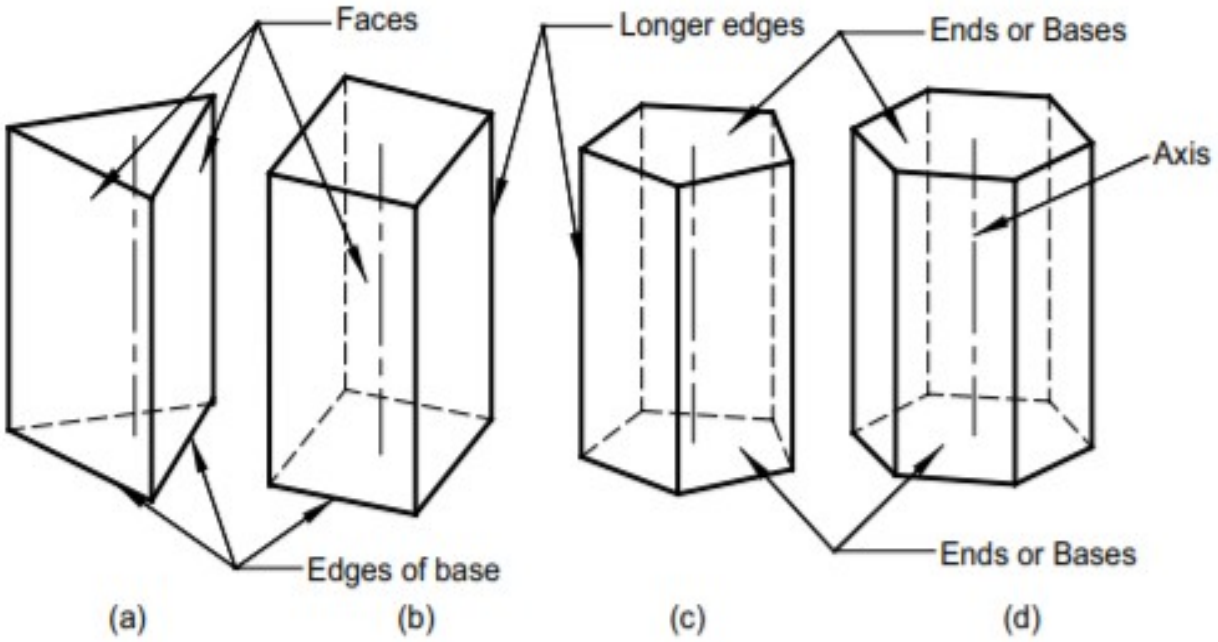


Fig. 11.3 *Prisms* **(a)** *Triangular* **(b)** *Square* **(c)** *Pentagonal* **(d)** *Hexagonal*

11.2.3 Pyramid

A pyramid is a polyhedron with n-sided polygonal base and lateral faces are triangles meeting at a point called the vertex or apex. An imaginary line that joins the apex with the centre of the base is known as the axis. *A right and regular pyramid* has a regular polygon base, axis perpendicular to the base and all the faces are equal isosceles triangles, as shown in Fig. 11.4. Pyramids are named according to the shape of their base, so a pyramid with a triangular base is called a *triangular pyramid*; a square base is called a *square pyramid* and so on. The centre of gravity of pyramids lies on the axis at one-fourth of its height from the base.

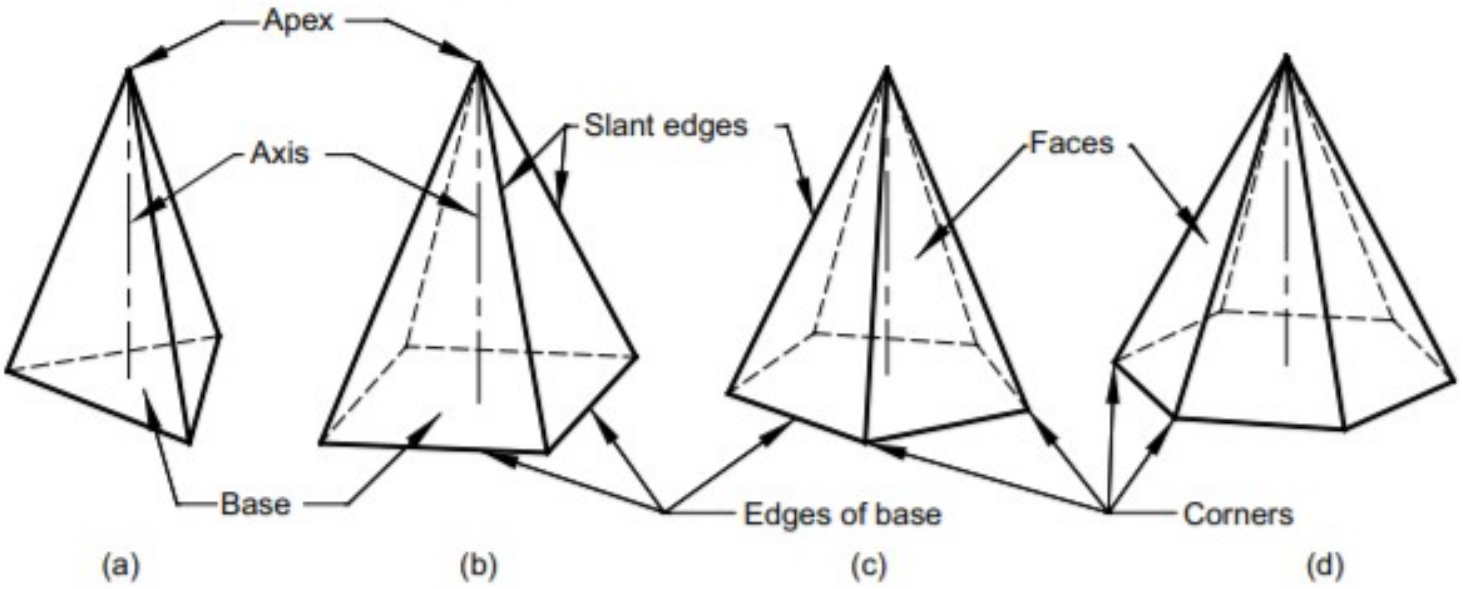


Fig. 11.4 *Pyramids* **(a)** *Triangular* **(b)** *Square* **(c)** *Pentagonal* **(d)** *Hexagonal*

Solid of Revolution

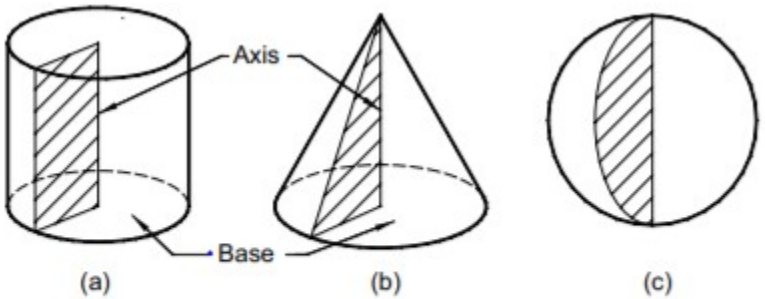


Fig. 11.5 *Solids of revolution* **(a)** *Cylinder* **(b)** *Cone* **(c)** *Sphere*

Oblique Solid

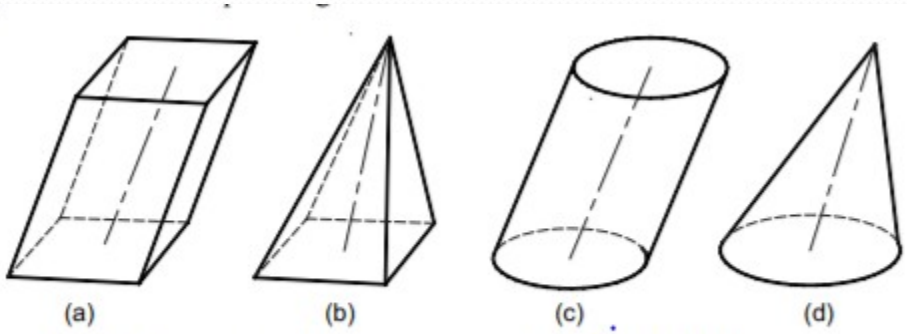


Fig. 11.6 *Oblique solids* **(a)** *Square prism* **(b)** *Square pyramid* **(c)** *Cylinder* **(d)** *Cone*

11.2.6 Frustum of Pyramid and Cone

When a regular pyramid or a cone is cut by a plane parallel to its base and the portion of the solid containing apex is removed, the remaining portion of the solid is called the *frustum* of that pyramid or cone, as shown in Fig. 11.7.

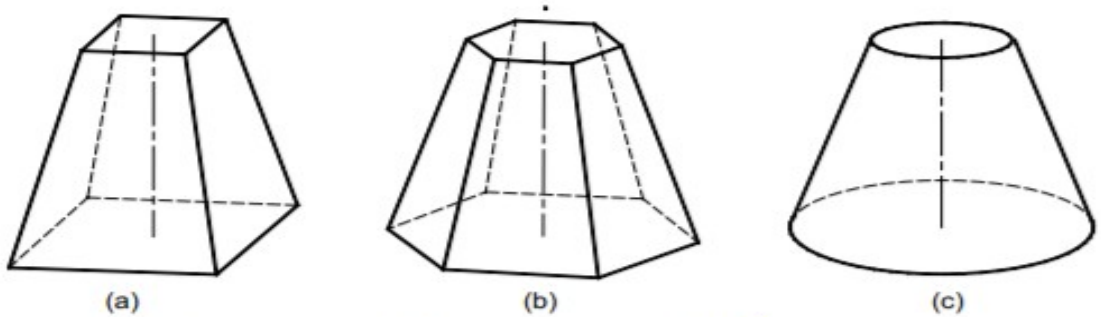
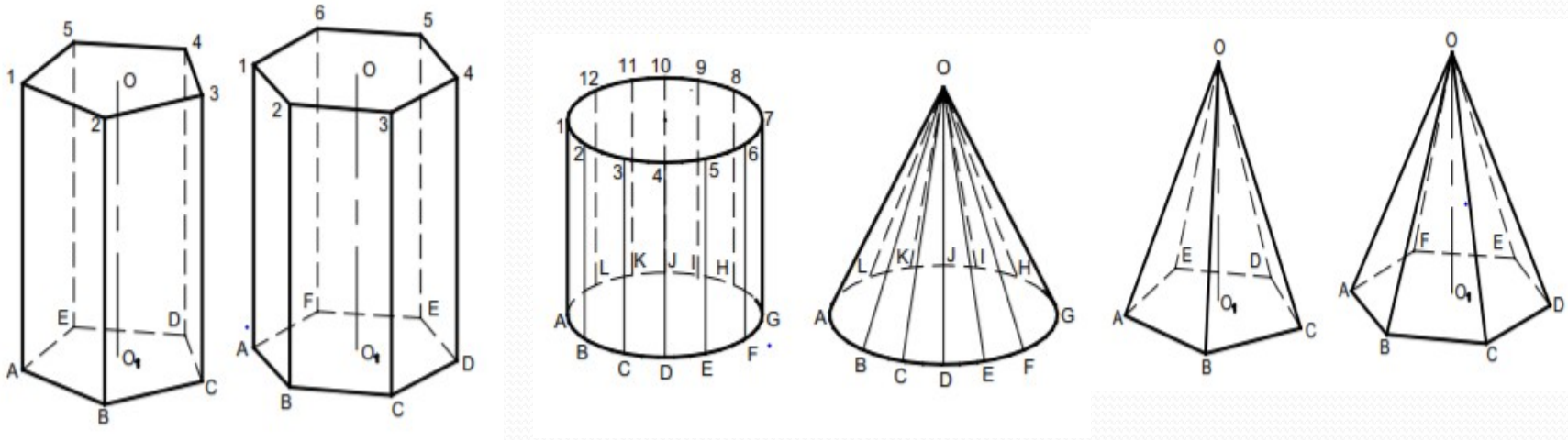


Fig. 11.7 *Frustums of* **(a)** *Square pyramid* **(b)** *Hexagonal pyramid* **(c)** *Cone*

RECOMMENDED METHOD OF LABELLING



ORIENTATION OF SOLID

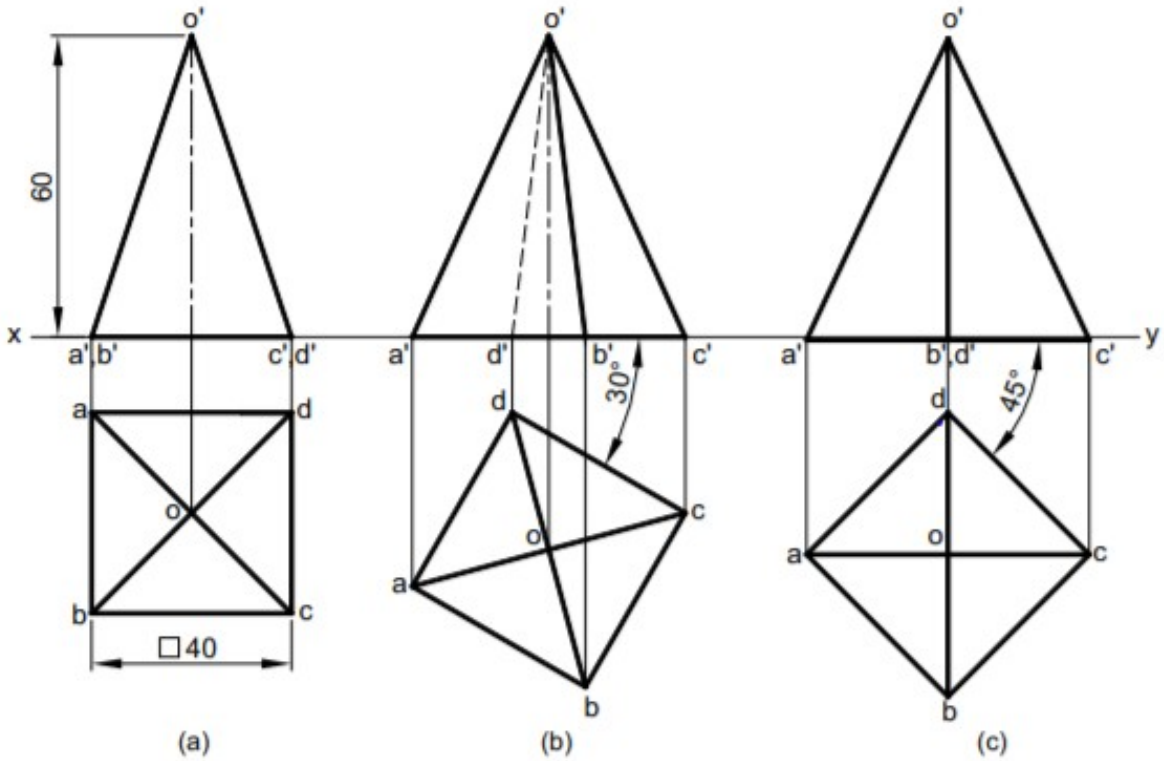
The solid may be in one of the following positions:

1. Axis perpendicular to the H.P.
2. Axis perpendicular to the V.P.
3. Axis parallel to both the H.P. and the V.P. (i.e., perpendicular to the profile plane)
4. Axis inclined to the H.P. and parallel to the V.P.
5. Axis inclined to the V.P. and parallel to the H.P.
6. Axis inclined to both the H.P. and the V.P.

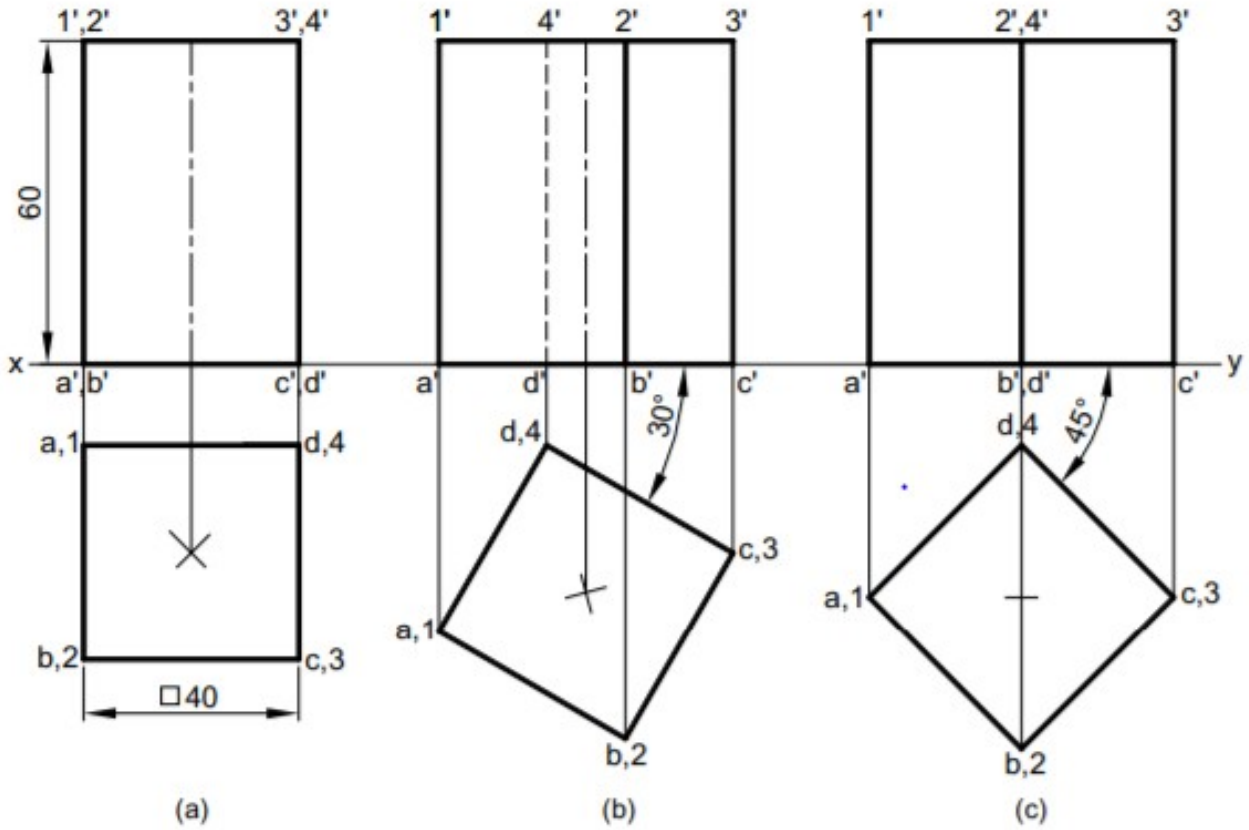
CASE-1

AXIS PERPENDICULAR TO H.P.

Problem 11.1 *A square pyramid of base side 40 mm and axis 60 mm is resting on its base on the H.P. Draw its projections when (a) a side of the base is parallel to the V.P., (b) a side of the base is inclined at 30° to the V.P., (c) all the sides of the base are equally inclined to the V.P.*

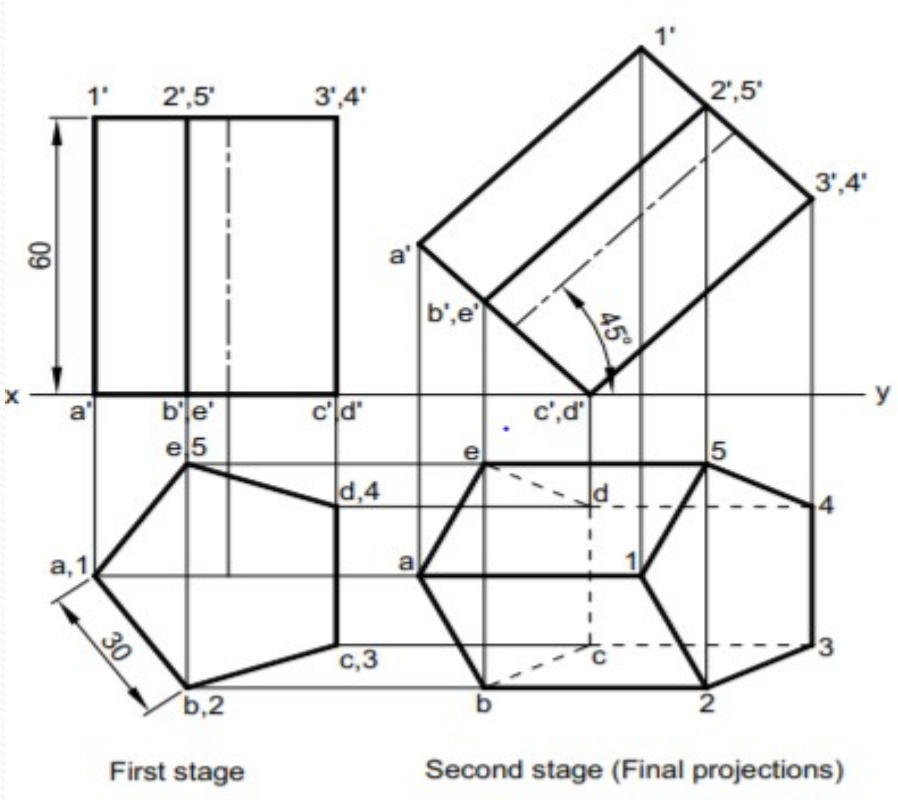


Problem 11.2 *A square prism of base side 40 mm and axis 60 mm is resting on its base on the ground. Draw its projections when (a) a face is perpendicular to the V.P., (b) a face is inclined at 30° to the V.P., (c) all the faces are equally inclined to the V.P.*

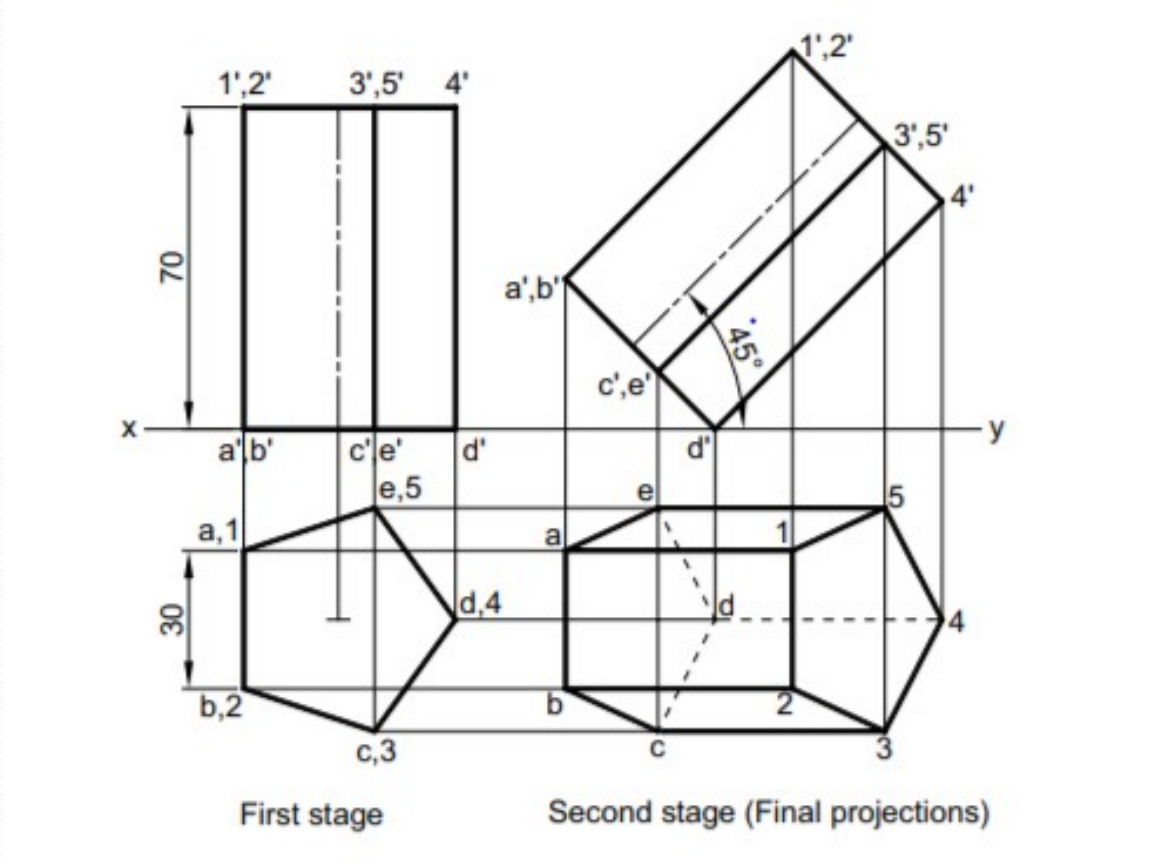


CASE-4 AXIS INCLINED TO H.P. AND PARALLEL TO V.P.

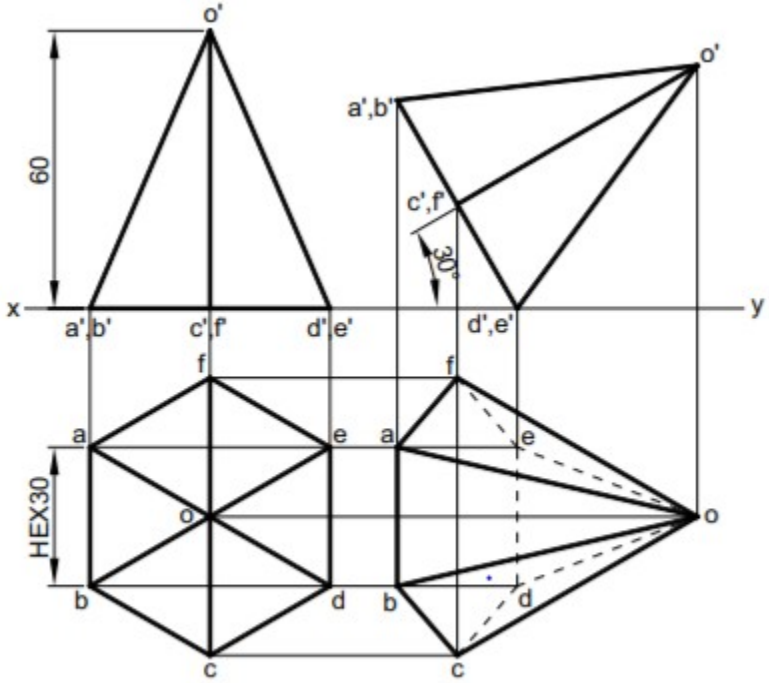
Problem 11.9 *A pentagonal prism of base edge 30 mm and axis 60 mm rests on an edge of its base in the H.P. Its axis is parallel to V.P. and inclined at 45° to the H.P. Draw its projections.*



Problem 11.10 *A pentagonal prism of base side 30 mm and axis 70 mm has a corners on the H.P. and the axis is inclined at 45° to the H.P. Draw its projection when the plane containing the resting corner and the axis is parallel to the V.P.*



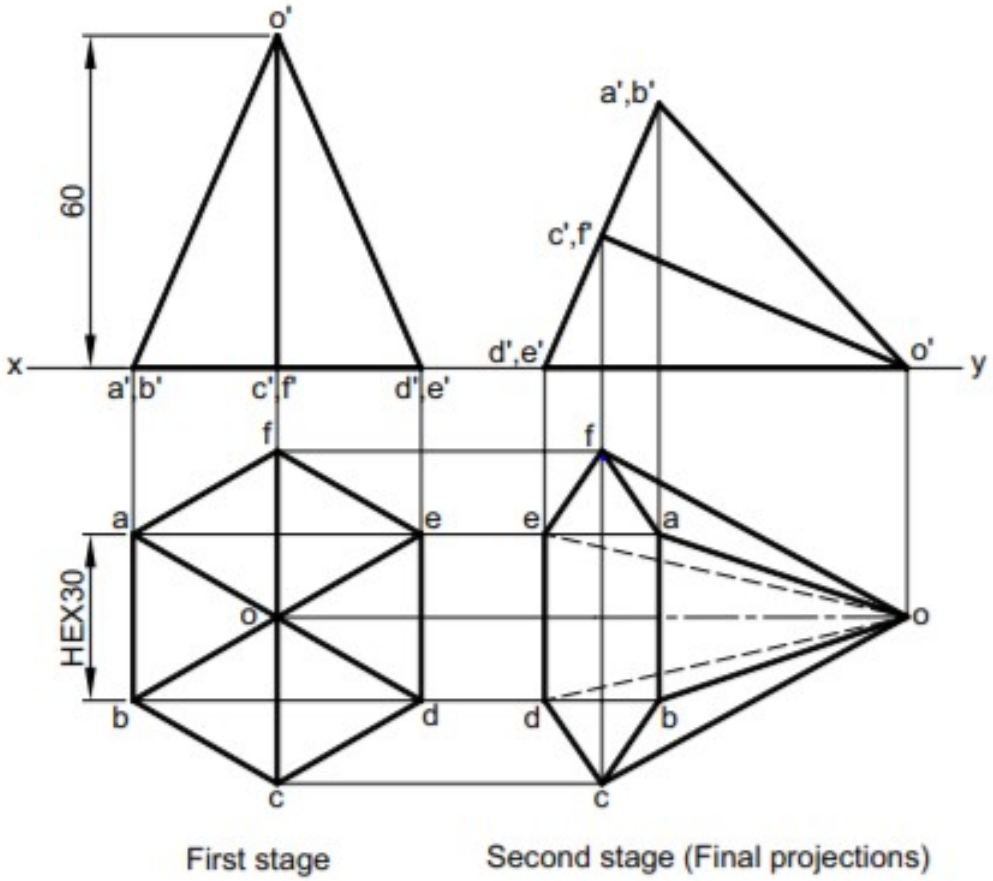
Problem 11.11 *A hexagonal pyramid of base side 30 mm and axis 60 mm has an edge of its base on the ground. Its axis is inclined at 30° to the ground and parallel to the V.P. Draw its projections.*



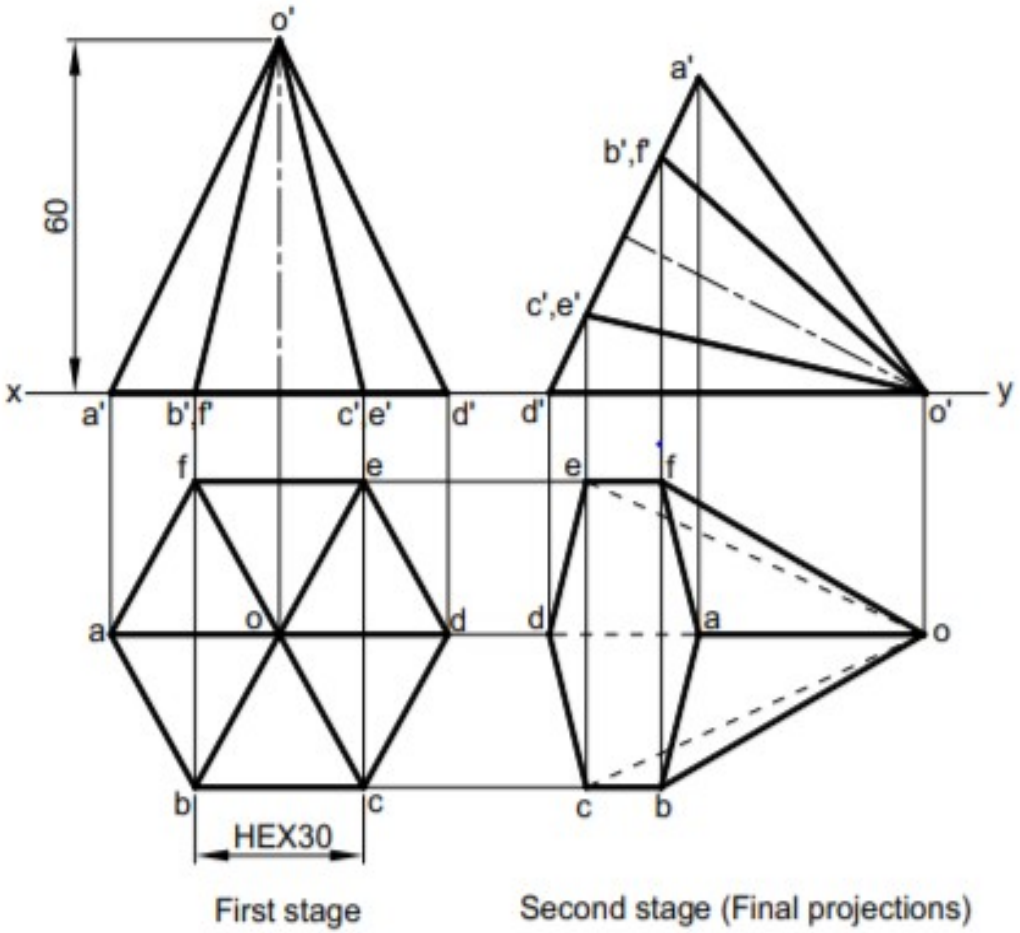
First stage

Second stage (Final projections)

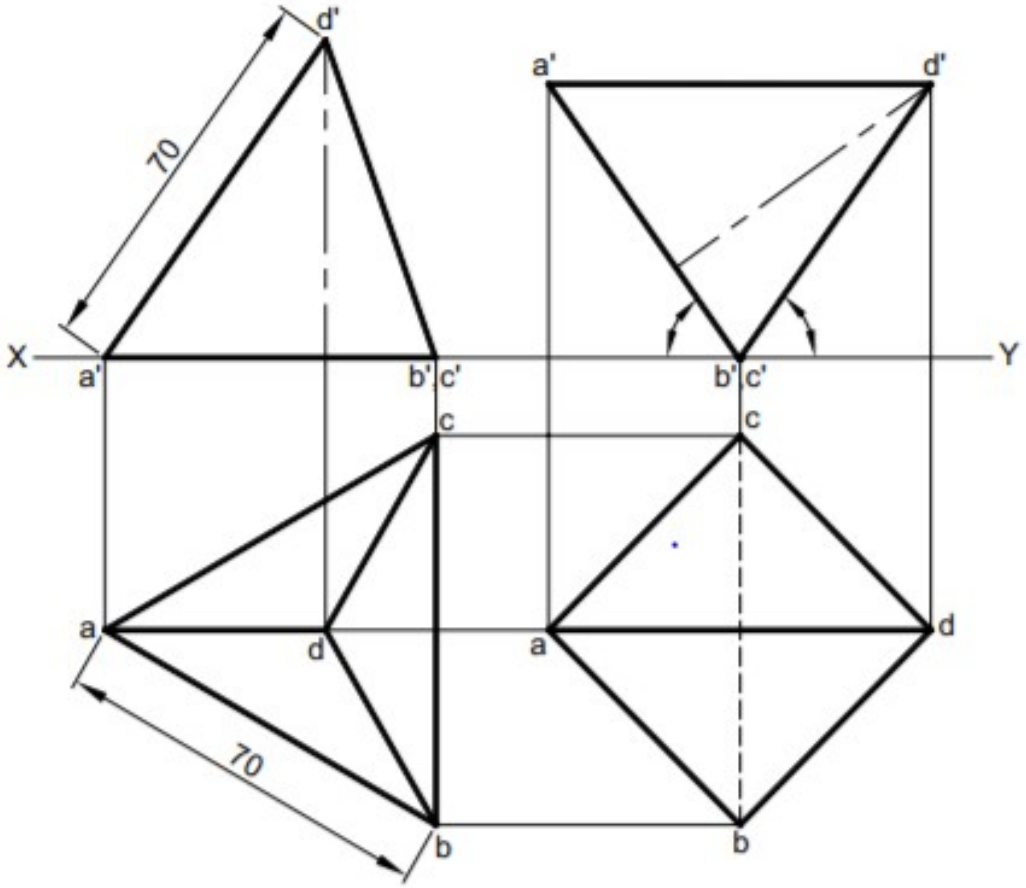
Problem 11.12 *A hexagonal pyramid of base edge 30 mm and axis 60 mm, has a triangular face on the ground and the axis parallel to the V.P. Draw its projections.*



Problem 11.13 *A hexagonal pyramid of base edge 30 mm and axis 60 mm, is lying on a slant edge on the ground with the axis parallel to the V.P. Draw its projections when the face containing the resting edge are equally inclined to the H.P.*



Problem 11.17 *A tetrahedron of edge 70 mm has an edge on the ground and the faces containing that edge are equally inclined to the H.P. Draw its projections when the edge lying on the ground is perpendicular to the V.P.*

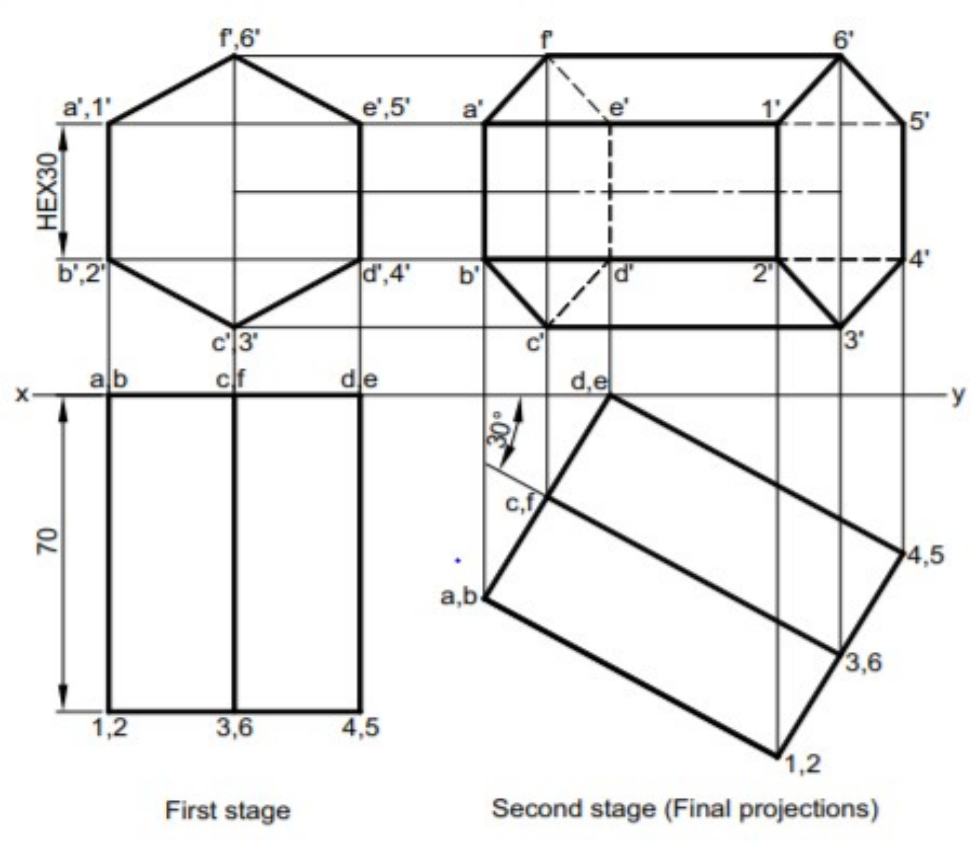


First stage

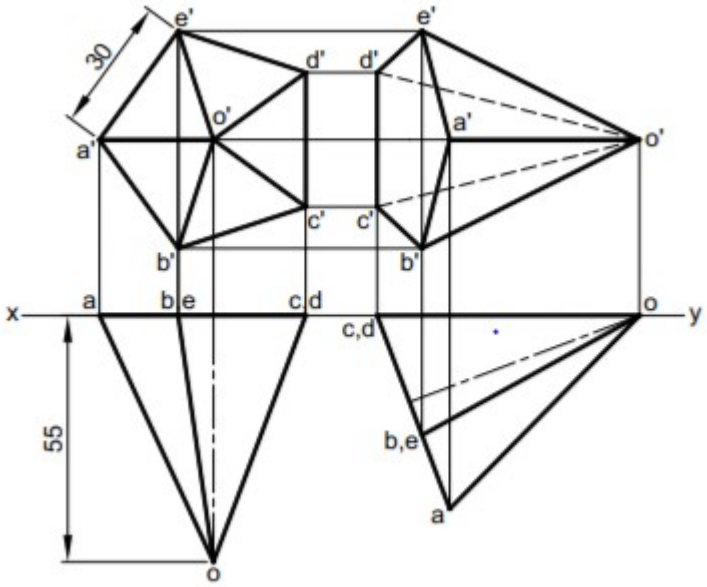
Second stage (Final projections)

CASE-5 AXIS INCLINED TO V.P. AND PARALLEL TO H.P.

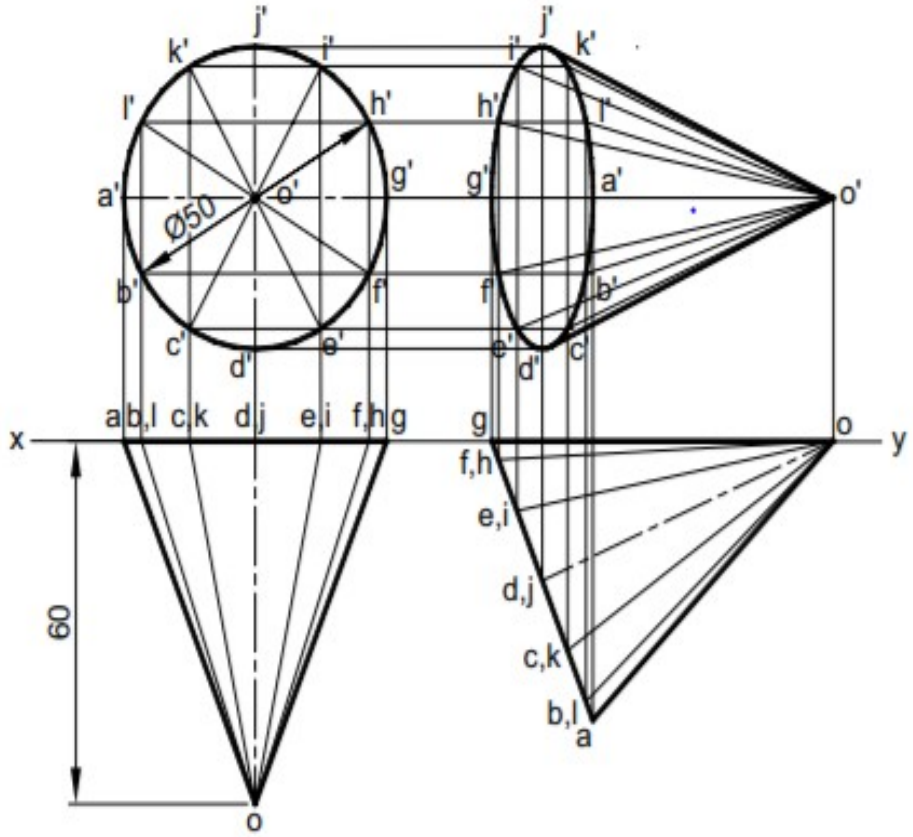
Problem 11.19 *A hexagonal prism of base edge 30 mm and axis 70 mm has an edge of its base in the V.P. such that the axis is inclined at 30° to the V.P. and parallel to the H.P. Draw its projections.*



Problem 11.20 *A pentagonal pyramid of base side 30 mm and axis 55 mm has a triangular face in the V.P. and the base edge contained by that triangular face is perpendicular to the H.P. Draw its projections.*



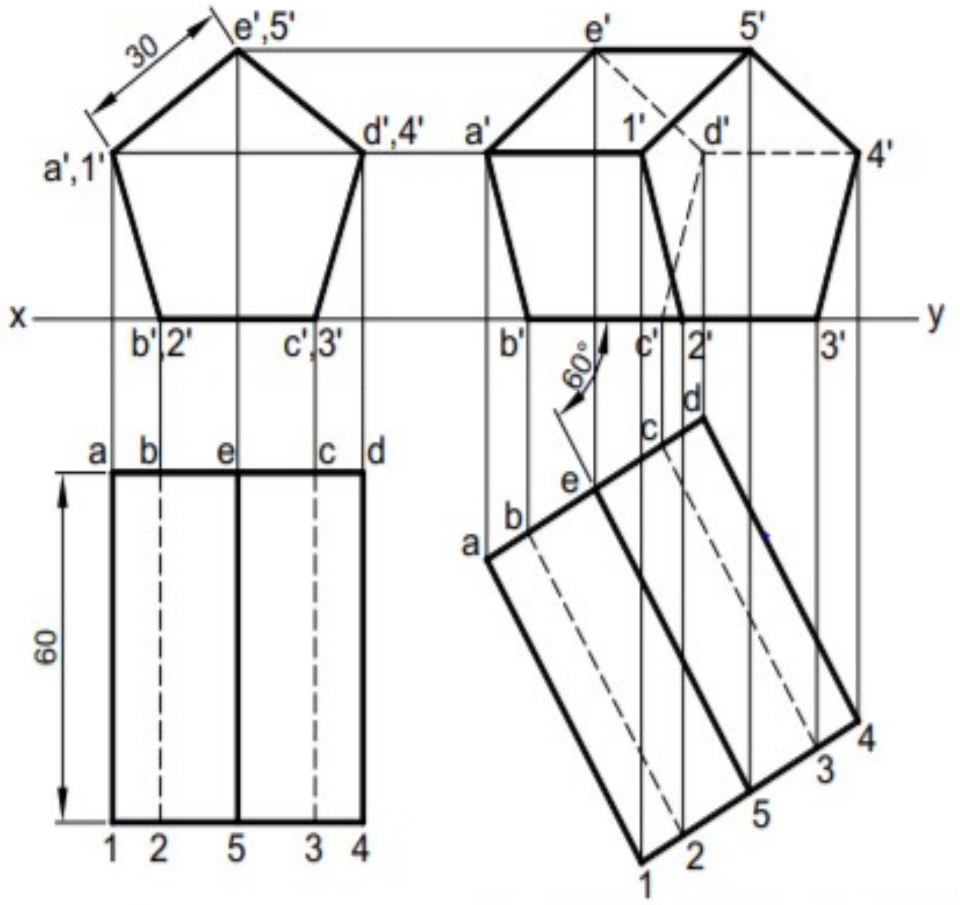
Problem 11.21 *A cone of base diameter 50 mm and axis 60 mm has a generator in the V.P. and the axis parallel to the H.P. Draw its projections.*



First stage

Second stage (Final projections)

Problem 11.22 *A pentagonal prism of base side 30 mm and axis 60 mm has one of its rectangular faces on the H.P. and the axis inclined at 60° to the V.P. Draw its projections.*



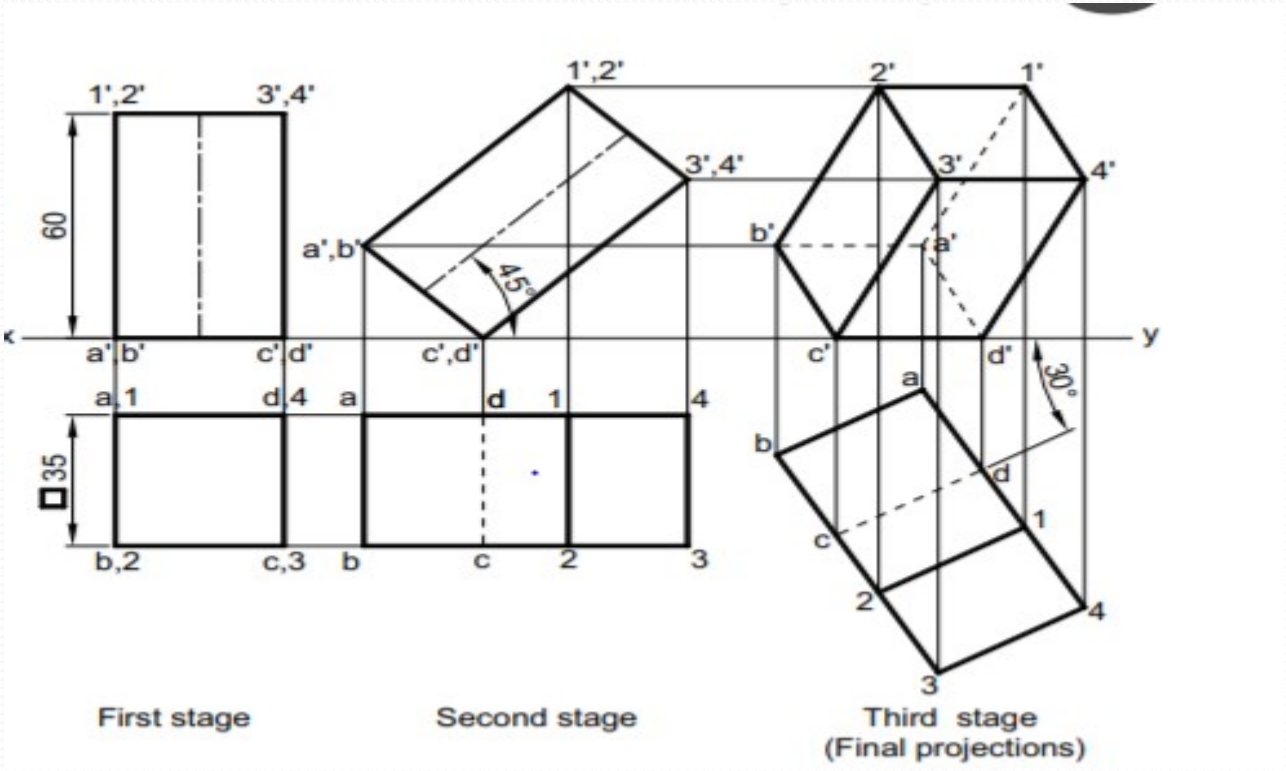
First stage

Second stage (Final projections)

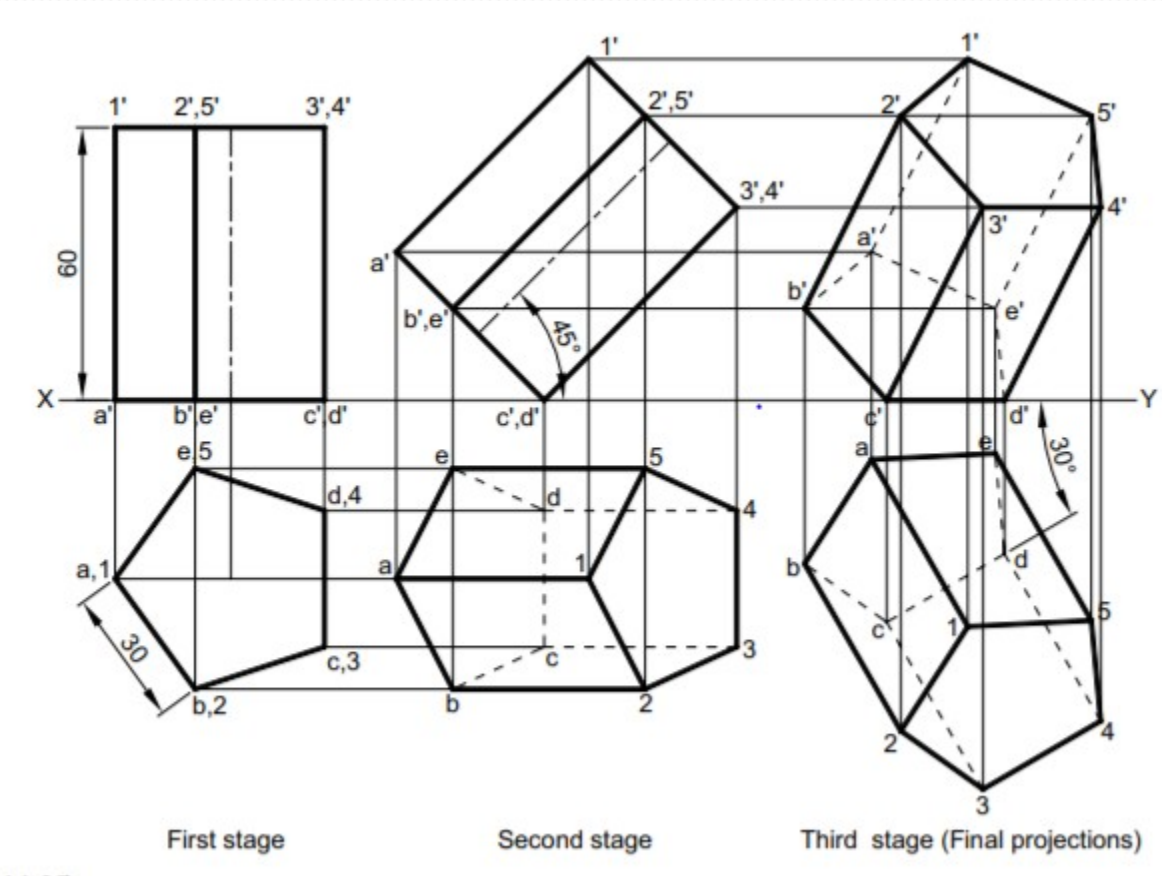
CASE-6

AXIS INCLINED TO BOTH H.P. AND V.P.

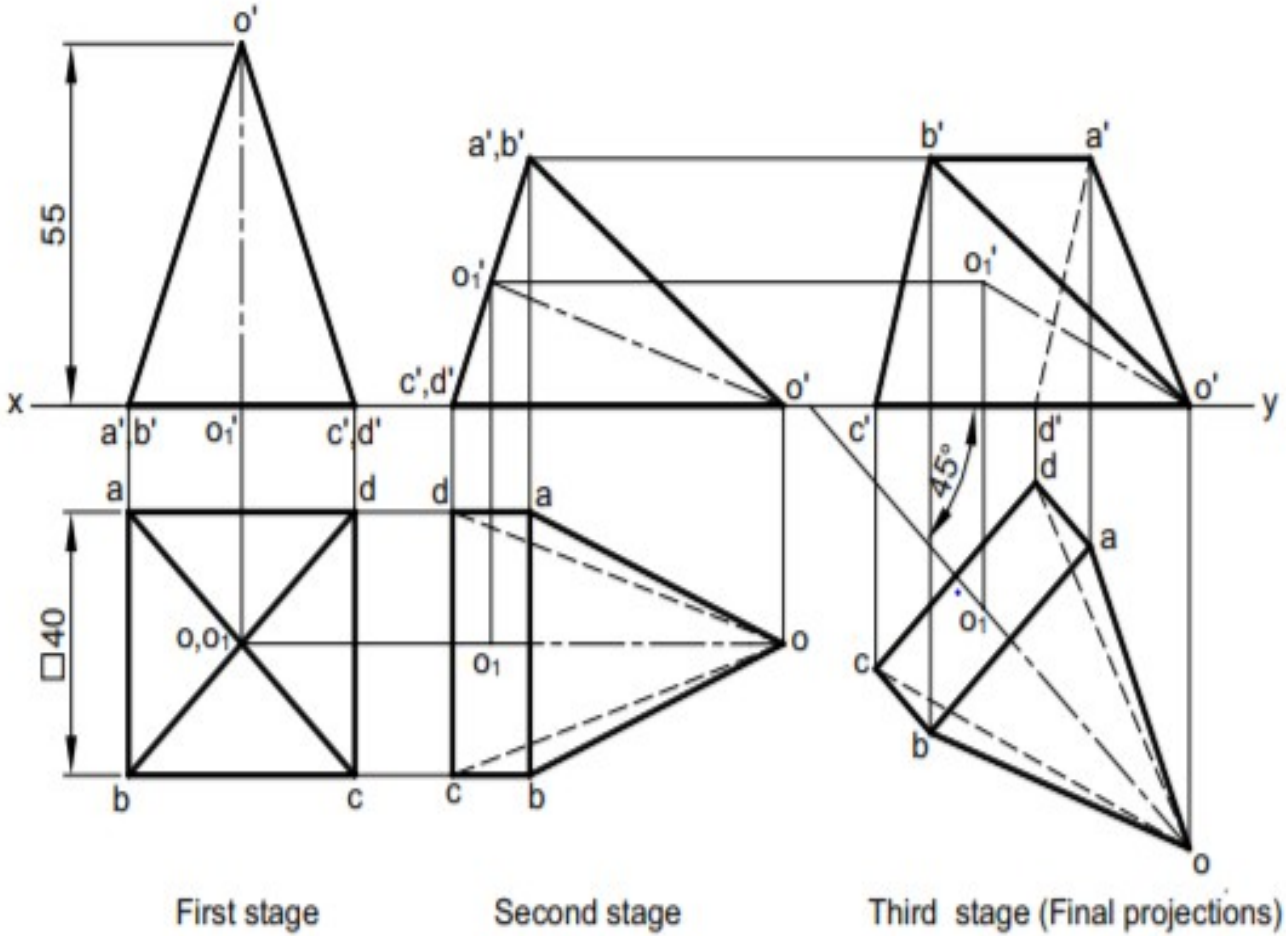
Problem 11.26 *A square prism of base edge 35 mm and axis 60 mm is resting on an edge of its base on the H.P. and the axis inclined at 45° to the H.P. If the edge resting on the H.P. is inclined at 30° to the V.P., draw its projections.*



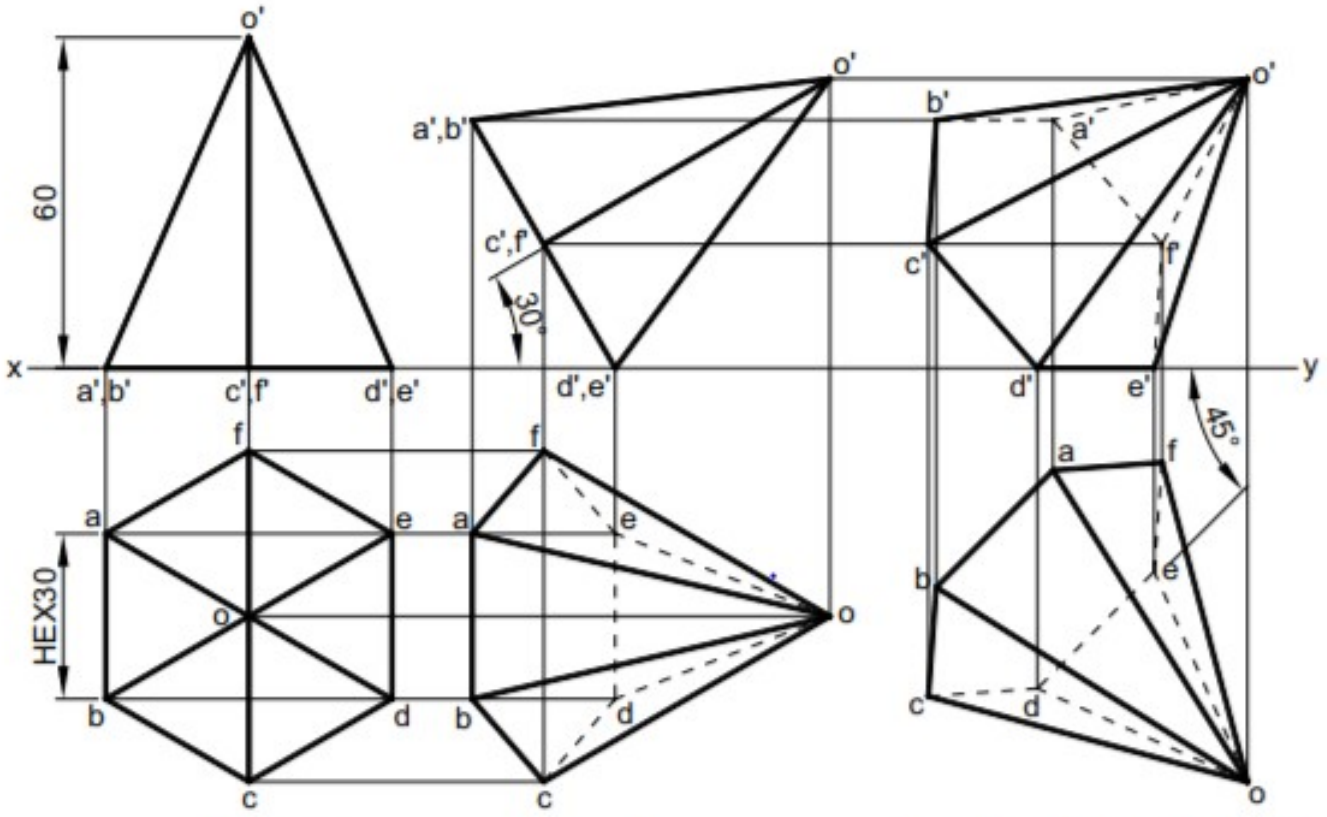
Problem 11.27 *A pentagonal prism of base side 30 mm and height 60 mm rests on one of its base side on the H.P. inclined at 30° to the V.P. Its axis is inclined at 45° to the H.P. Draw its projections.*



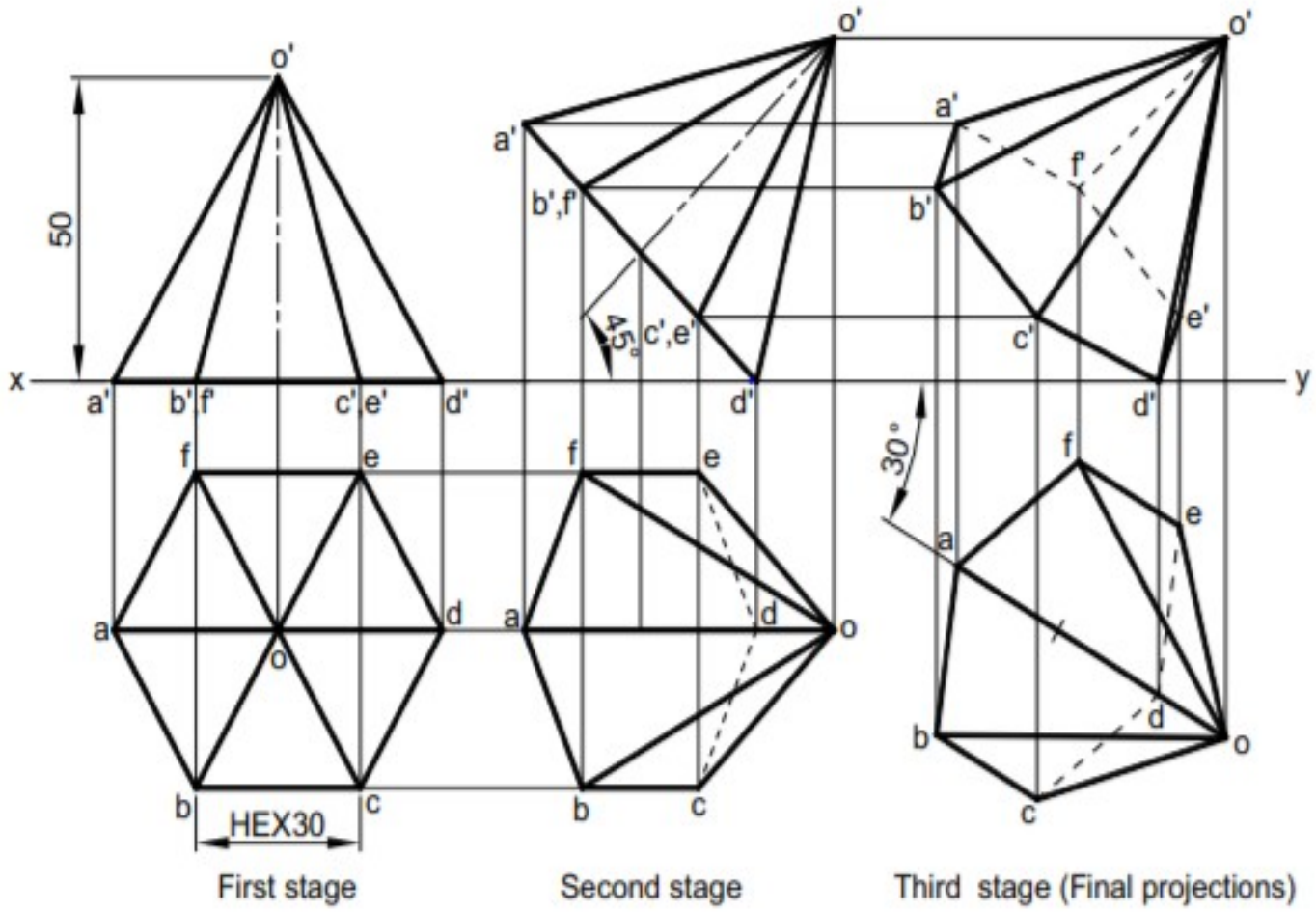
Problem 11.28 *A square pyramid of base side 40 mm and axis 55 mm is resting on one of its triangular faces on the H.P. A vertical plane containing the axis is inclined at 45° to the V.P. Draw its projections.*



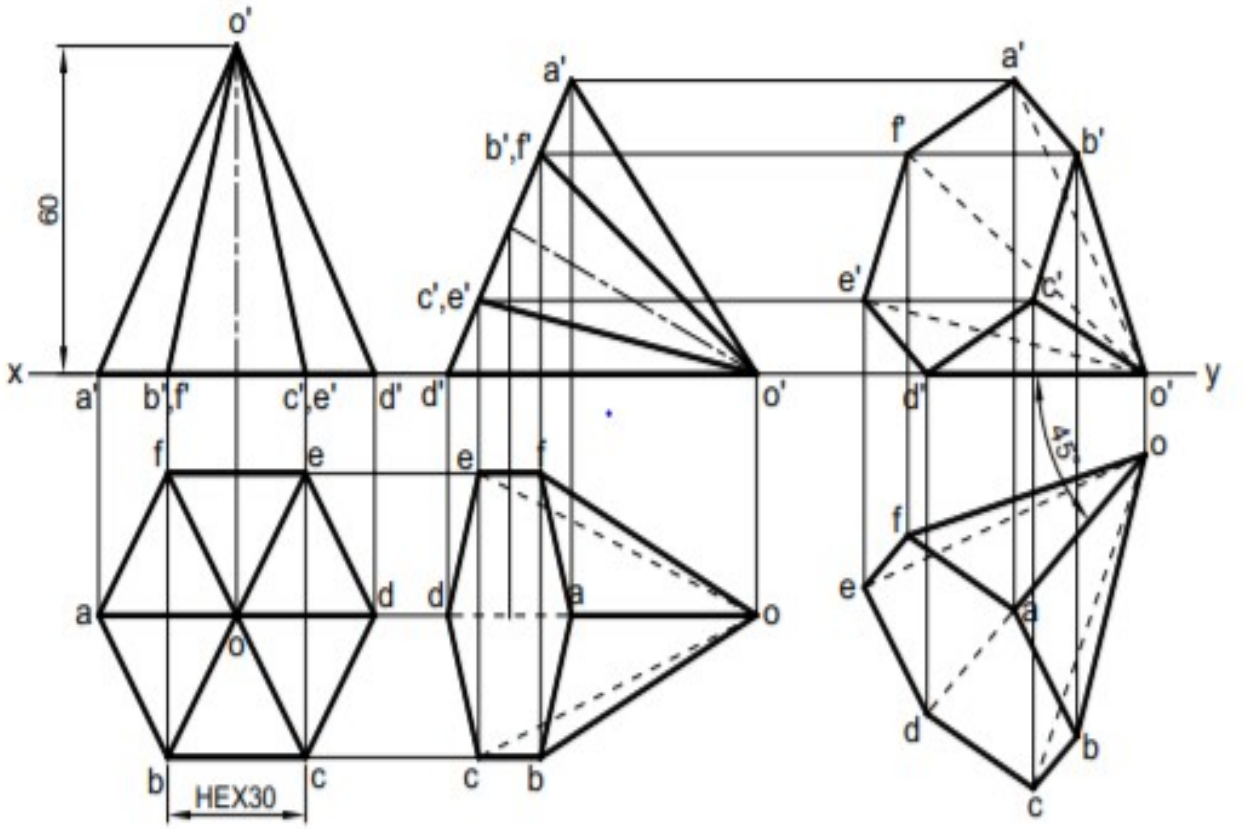
Problem 11.29 *A hexagonal pyramid of base side 30 mm and axis 60 mm, has an edge of its base on the ground inclined at 45° to the V.P. and the axis is inclined at 30° to the H.P. Draw its projections.*



Problem 11.30 *A hexagonal pyramid of base side 30 mm and axis 50 mm, rests on one of its base corners on the ground with axis inclined at 45° to the H.P. Draw its projections when a vertical plane containing the axis and the corner that lies in the H.P. is inclined at 30° to the V.P.*

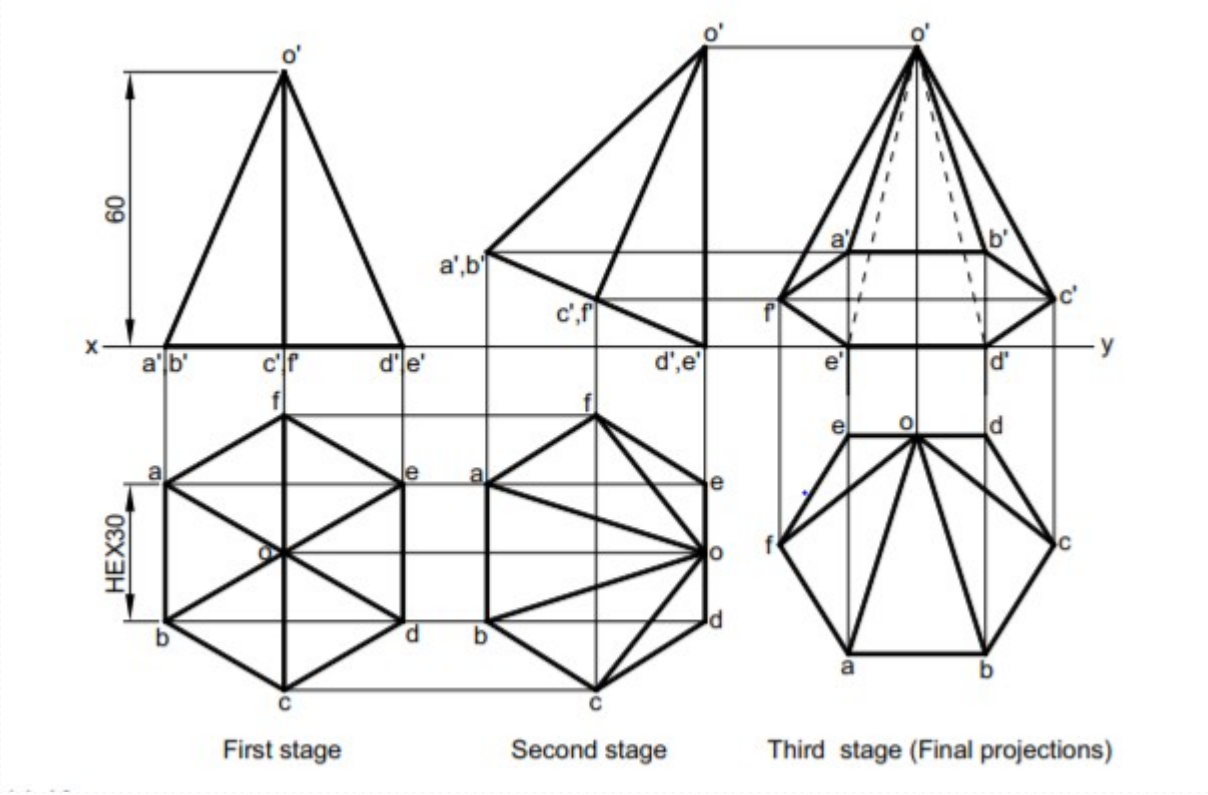


Problem 11.31 *A hexagonal pyramid of base side 30 mm and axis 60 mm has one of its slant edges on the H.P. and inclined at 45° to the V.P. Draw its projections when the base is visible.*

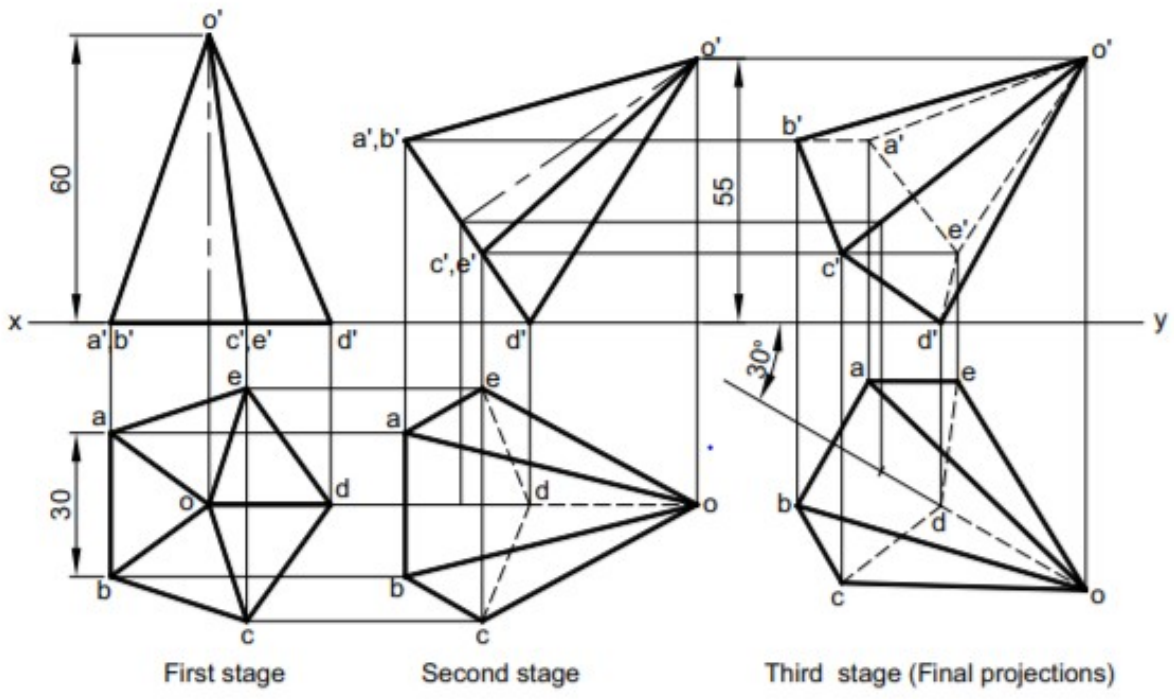


First stage Second stage Third stage (Final projections)

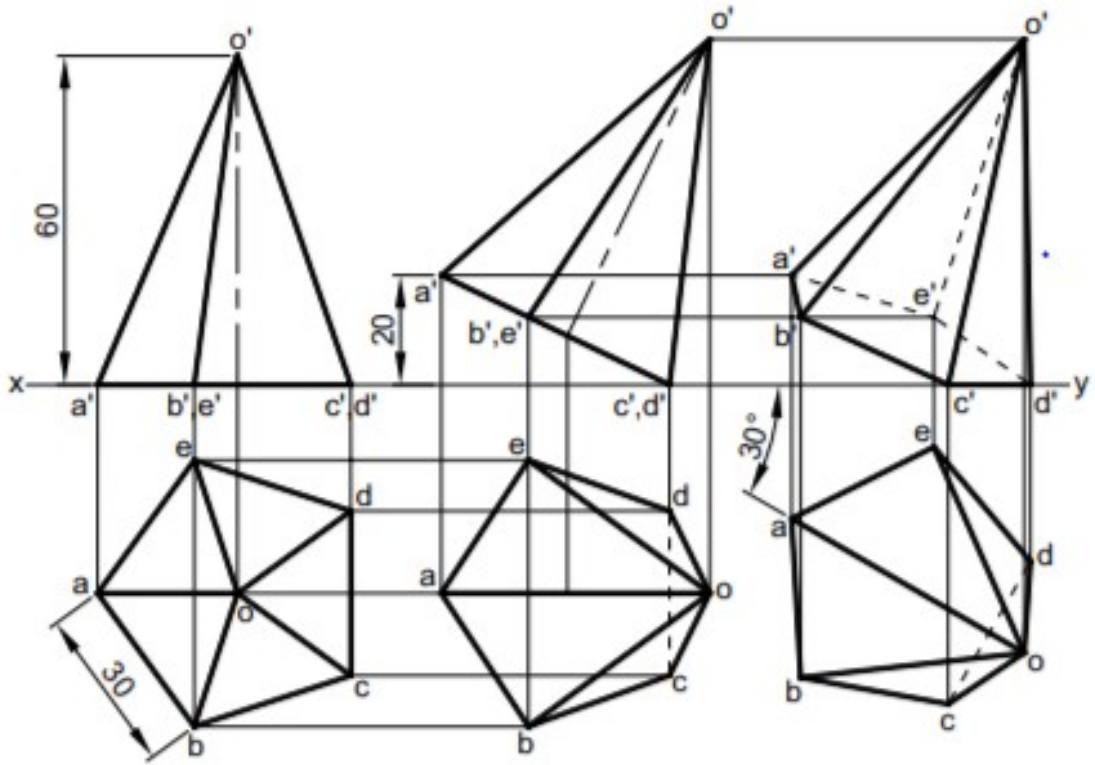
Problem 11.32 *A hexagonal pyramid of base side 30 mm and axis 60 mm rests on an edge of base on the H.P. with the triangular face containing that edge perpendicular to the H.P. and parallel to the V.P. Draw its projections so that the base is visible.*



Problem 11.33 *A pentagonal pyramid of base side 30 mm and axis 60 mm rests on a corner of its base on the H.P. such that its apex is 55 mm above the ground. A vertical plane containing the corner of the base that lies on the H.P. and the axis is inclined at 30° to the V.P. Draw its projections.*



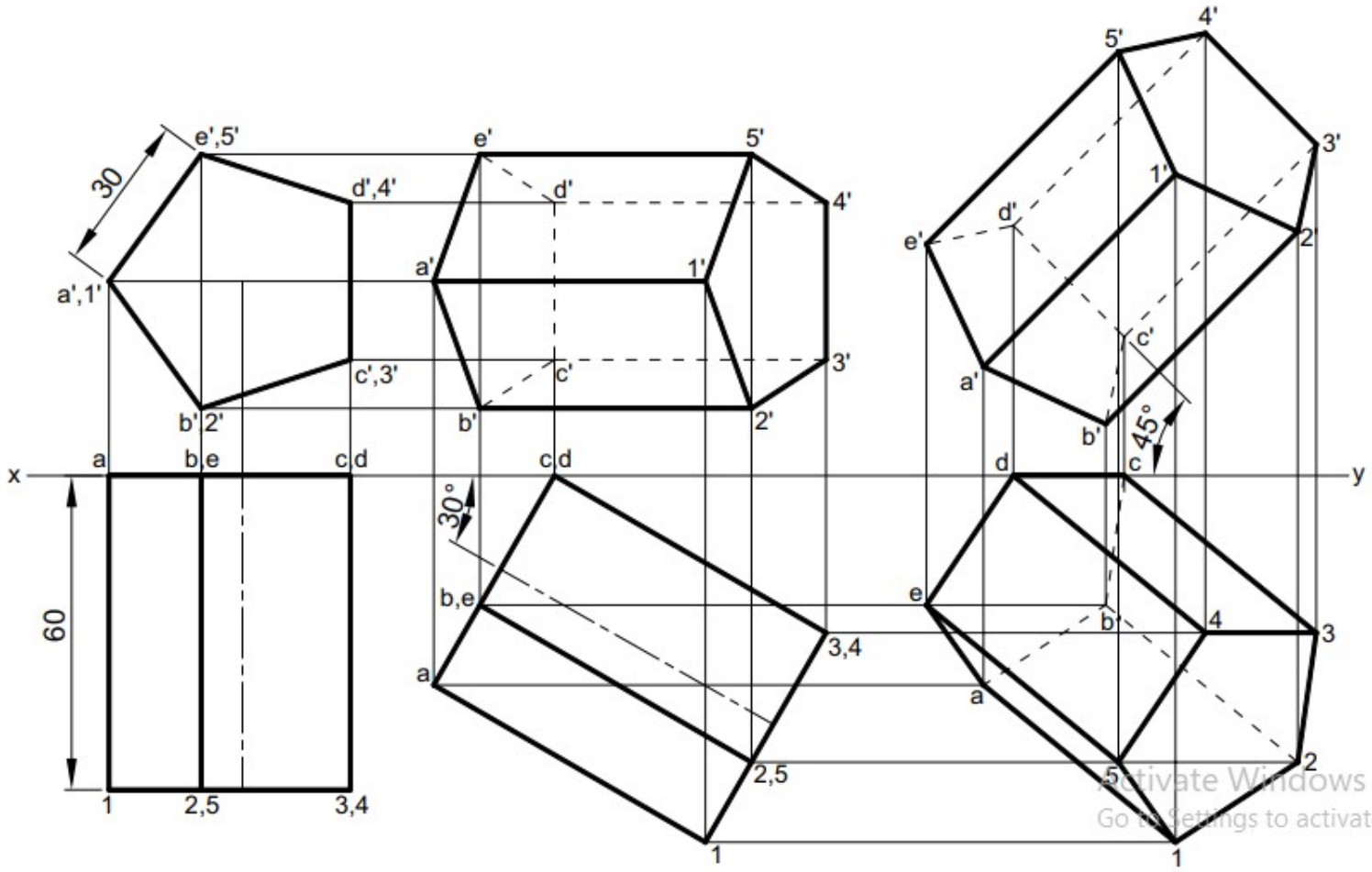
Problem 11.34 *A pentagonal pyramid of base side 30 mm and axis 60 mm rests on an edge of its base on the ground so that the highest point of the base is 20 mm above the ground. Draw its projections when a vertical plane containing the axis is inclined at 30° to the V.P.*



First stage Second stage Third stage (Final projections)

Zoom out (Ctrl+Minus key)

Problem 11.43 *A pentagonal prism of base side 30 mm and axis 60 mm has an edge of its base in the V.P. and inclined at 45° to the H.P. Its axis is inclined at 30° to the V.P. Draw its projections.*



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Go to Settings to activate Windows.